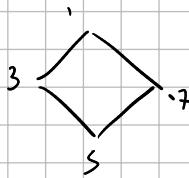
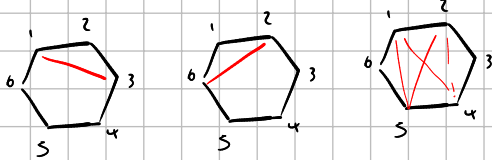


• triangles are not killed



things that have a triangle $\langle 12k \rangle$ are not killed

but not the set anything that has a triangle $\langle i, i+1, k \rangle$ is not killed

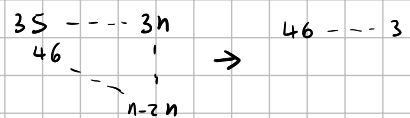
D subsets

$$F_{13} = \{(i, j) \mid 3 \leq i \leq n-2, i+2 \leq j \leq n\}$$

* bombing properties (subset of F_{13})

$\nearrow$ $k=1$ choose $S \in F_{13}^{n-4}$

$$n=6 \quad F = \{35, 36, 46\}$$



ex:

$$B = \frac{a_1}{x_{35} x_{36} x_{13}} + \frac{a_2}{x_{33} x_{36}^2} + \frac{a_3}{x_{33} x_{46} x_{24}} + \frac{a_4}{x_{35} x_{36} x_{24}}$$

$$= \frac{1}{x_{35} x_{36}} \left(\frac{a_1}{x_{13}} + \frac{a_4}{x_{24}} \right) + \frac{a_2}{x_{33} x_{36}^2} + \frac{a_3}{x_{33} x_{46} x_{24}}$$

"0"
"0"
"0"

$$\rightarrow \sum_{A \in \overline{F}_{13}} \frac{a_{AS}}{x_A} = 0 \quad \leftarrow \text{since D-subset}$$

$$a_{135} = a_{2n5}$$

$$a_{AS} = 0 \text{ if } A \neq (13), (2n)$$

some are going to be zero if S has crossings

← already know by step 1

new

$k=2$

$$\sum_{A, A_2 \in \overline{F}_{13}^2} \frac{a_{A, A_2 S}}{x_A x_{A_2}} = 0$$

some thys are zero by step 1 (A, A_2 are not $(13)(2n)$ or $(1k)(2k)$ at the same time)

what's left:

$\leftarrow l = 4, \dots, n-1$

$$\underline{X_{13}^2}, \underline{X_{2n}^2}, \underline{X_{13} X_{2n}}, \underline{X_{13} X_{1l}}, \underline{X_{13} X_{2l}}, \underline{X_{2n} X_{1l}}, \underline{X_{2n} X_{2l}}, \underline{X_{1l} X_{2l}}$$

$$X_{2l} = X_{1l} - X_{13} \quad l = 4, \dots, n-1$$

$$X_{2n} = -X_{13}$$

common denominator $X_{13}^2 X_{2n}^2 X_{14} X_{15} \dots X_{1,n-1} X_{24} X_{25} \dots X_{2,n-1}$

terms in numerator: $a_1 \underline{X_{2n}^2 \prod_l X_{1l} X_{2l}} + a_2 \underline{X_{13}^2 \prod_l X_{1l} X_{2l}} + a_3 \underline{X_{13} X_{2n} \prod_l X_{1l} X_{2l}}$

for each $q = 4, \dots, n-1$

$$\left\{ \begin{aligned} & b_{1q} \underline{X_{13} X_{2n}^2 X_{2q} \prod_{l \neq q} X_{1l} X_{2l}} + b_{2q} \underline{X_{13} X_{2n}^2 X_{1q} \prod_{l \neq q} X_{1l} X_{2l}} + \\ & b_{3q} \underline{X_{13}^2 X_{2n} X_{2q} \prod_{l \neq q} X_{1l} X_{2l}} + b_{4q} \underline{X_{13}^2 X_{2n} X_{1q} \prod_{l \neq q} X_{1l} X_{2l}} + \\ & b_{5q} \underline{X_{13}^2 X_{2n}^2 \prod_{l \neq q} X_{1l} X_{2l}} \end{aligned} \right.$$

set some stuff to zero in a way that leaves only 2 terms alive
and send 13 or 2n to zero.

try to send $X_{1q} \rightarrow 0$: 3 survive:

$$\begin{aligned} a_1 \underline{X_{13} X_{2n}^2 X_{2q} \prod_{l \neq q} X_{1l} X_{2l}} &\rightarrow -X_{13}^4 \prod_{l \neq q} X_{1l} X_{2l} \\ a_2 \underline{X_{13}^2 X_{2n} X_{2q} \prod_{l \neq q} X_{1l} X_{2l}} &\rightarrow +X_{13}^4 \prod_{l \neq q} X_{1l} X_{2l} \\ a_3 \underline{X_{13}^2 X_{2n}^2 \prod_{l \neq q} X_{1l} X_{2l}} & \quad X_{13}^4 \prod_{l \neq q} X_{1l} X_{2l} \end{aligned}$$

$$b_{1q} = b_{3q} + b_{5q}$$

send $X_{2q} \rightarrow 0$:

$$\begin{aligned} X_{1q} &\rightarrow X_{13} \\ X_{2n} &\rightarrow -X_{13} \end{aligned}$$

$$\begin{aligned} & b_{2q} \underline{X_{13} X_{2n}^2 X_{1q} \prod_{l \neq q} X_{1l} X_{2l}} \\ & b_{4q} \underline{X_{13}^2 X_{2n} X_{1q} \prod_{l \neq q} X_{1l} X_{2l}} \\ & b_{5q} \underline{X_{13}^2 X_{2n}^2 \prod_{l \neq q} X_{1l} X_{2l}} \end{aligned}$$

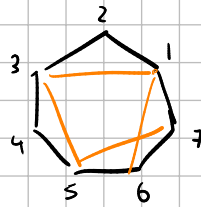
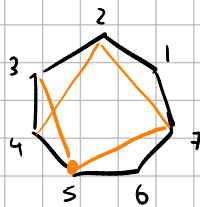
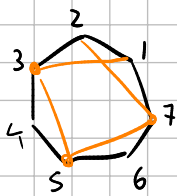
$$X_{13}^4$$

$$-X_{13}^4$$

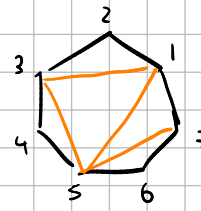
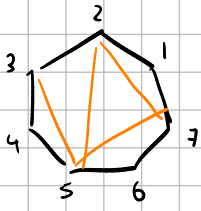
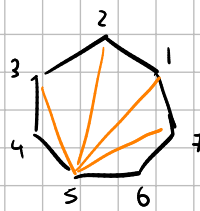
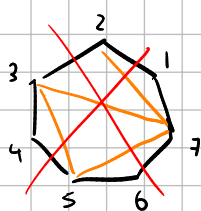
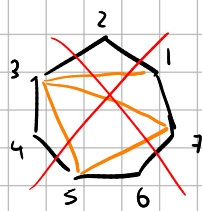
$$X_{13}^4$$

$$b_{5q} + b_{2q} = b_{4q}$$

$$n=7$$



$$\begin{array}{ccc} 35 & 36 & 37 \\ & 46 & 47 \\ & & 57 \end{array}$$



→ not allowed because they have 1 boundary neighbor (red 2)

$$k=2$$

choose our $n-5$ thys from $\overline{F}$ as $(35), (57)$

$$\text{then } \frac{b_1}{X_{13} X_{27}} + \frac{b_2}{X_{24} X_{27}} + \frac{b_3}{X_{16} X_{13}} + \frac{a_1}{X_{15} X_{25}} + \frac{a_2}{X_{27} X_{25}} + \frac{a_3}{X_{15} X_{13}} = 0$$

Common denominator: $X_{13} X_{15} X_{16} X_{24} X_{25} X_{27}$

numerator: $0 = b_1 X_{15} X_{16} X_{24} X_{25} X_{27} + b_2 X_{13} X_{15} X_{16} X_{25} + b_3 X_{15} X_{24} X_{25} X_{27} + a_1 X_{13} X_{16} X_{24} X_{27} + a_2 X_{13} X_{15} X_{16} X_{24} + a_3 X_{16} X_{24} X_{25} X_{27}$

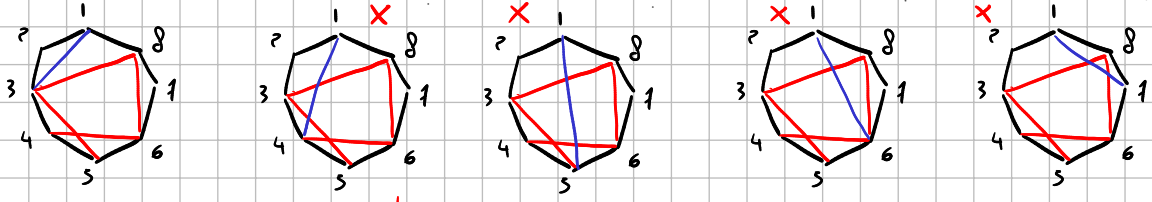
and $16 \Rightarrow 0 \Rightarrow b_3 = 0$

$15 \Rightarrow 0 : a_1 X_{13} X_{16} X_{24} X_{27} + a_3 X_{16} X_{24} X_{25} X_{27} = 0 \quad (a_1 = a_3)$

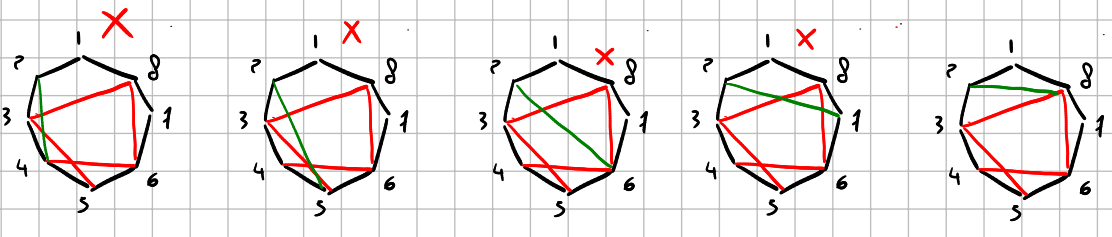
$24 \Rightarrow 0 \Rightarrow b_2 = 0$

$25 \Rightarrow 0 : a_1 = a_2$

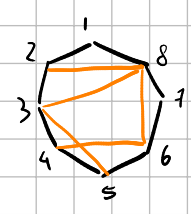
$n=8$



→ killed by step 1
(not every edge is part of a triangle)

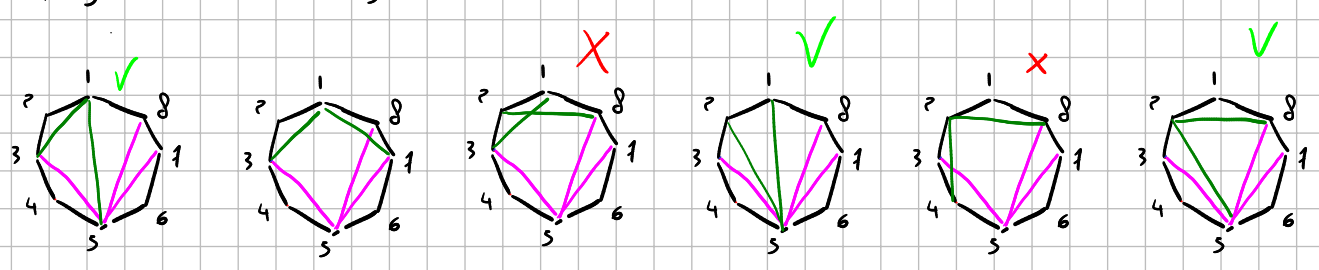
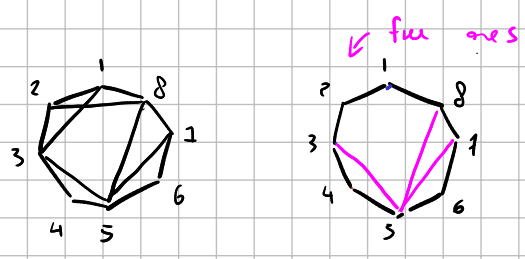


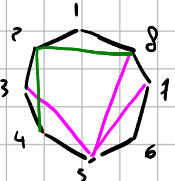
- 35 36 34 38
- 46 47 48
- 57 58
- 68



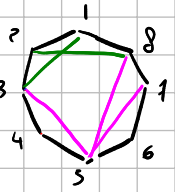
Same Geometric
(proved earlier)

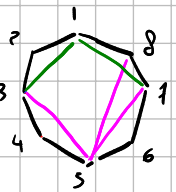
let's rotate so the same shape has 2 boundary projectors.



send $X_{24} \rightarrow 0$ tells us that  is killed (+ cyclic rotations)

- send $X_{13} \rightarrow 0$
- $X_{24} = X_{14}$
 - $X_{25} = X_{15}$
 - $X_{26} = X_{16}$
 - $X_{27} = X_{17}$
 - $X_{28} = 0$

number coeff. of  $\propto X_{15} X_{17} X_{24} X_{25}$
→ is killed

send $17 \rightarrow 0$ kills 

insights at 2 body props. not lev X_{13}^2

because can't have any other body propagators and have $n-5$ left

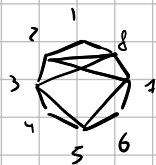


Q: how many chords are needed to make a pseudo-triangulation of an n -gon?

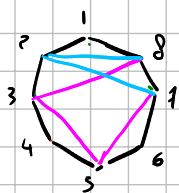
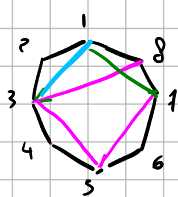
↳ a collection of chords where every edge is part of a triangle.

does not work

$\Gamma_{n/2}$



2 body prop $(28, 27)$



$$\frac{a_1}{X_{13} X_{17}} + \frac{a_2}{X_{27} X_{28}} = 0$$

$$\Rightarrow a_1 X_{27} X_{28} + a_2 X_{13} X_{17} = 0$$

both zero